

Experimental observation of plasmon–polariton waves supported by a thin metal film of finite width

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What are believed to be the first experimental observations of the existence of long-range plasmon–polariton waves, guided by a thin metal film of finite width, are presented. A waveguide composed of an 8- μm -wide, 20-nm-thick, 3.5-mm-long Au metal film embedded in SiO_2 was successfully excited at a free-space wavelength of 1.55 μm in an end-fire experiment. The theoretical nature of the phenomenon is described, and experimental observations of field confinement provided by this metal waveguide are presented in detail.

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At optical wavelengths, metals are characterized by a dielectric constant with a negative real part and can be modeled as a free-electron gas or plasma. Knowing that surface plasmon–polaritons can be guided at an interface between two media with real parts of permittivity of opposite signs,¹ it becomes clear that a metal embedded in a dielectric can support plasmon–polariton waves when the metal is properly excited.

Surface plasmon–polaritons attracted much interest in the late 1970's and the early 1980's.^{2–4} Since surface plasmon–polaritons propagate at the metal–dielectric interface, they are sensitive to surface irregularities and thus provide a means of assessing the quality of a metal surface. Until now, research concentrated only on thin metal films of infinite width supported by an optically transparent dielectric for possible use in surface spectroscopy. Many intriguing properties of these films were discovered and reported over the years. Theoretical analysis revealed that only TM modes were supported by these thin metal films. Two bound propagating modes were identified for a symmetric structure consisting of a thin metal film of infinite width embedded in a homogeneous dielectric¹: the symmetric bound (s_b) and the asymmetric bound (a_b) modes, identified as such for their respective field patterns.

The symmetric mode of slab films is of great interest, since its field configuration lends itself to easy optical excitation and it exhibits dispersion with metal thickness, such that a significant decrease in attenuation is expected for a diminishing film thickness. Indeed, this was experimentally verified.⁵ Attenuated total reflection measurements revealed an exponential increase in propagation distance (or equivalently, a decrease in attenuation) as the thickness of the metal was reduced. End-fire excitation of slab waveguides was proposed⁶ but to our knowledge never demonstrated. The asymmetric mode, which is also bound to thin metal films of infinite width, is less attractive than the symmetric mode because it is less easily excited by use

of an end-fire configuration and is characterized by increasing attenuation for decreasing metal thickness.

Recently, it was discovered through optical modal analysis that thin metal films of finite width, as shown in the inset of Fig. 1, support plasmon–polariton modes.^{7,8} The properties of these metallic waveguides are similar to those of infinitely wide structures, with the additional benefit of providing two-dimensional field confinement. Once again, symmetric and asymmetric modes can propagate, but owing to the inherent two-dimensional confinement, these modes exhibit field variations in both the lateral and the vertical

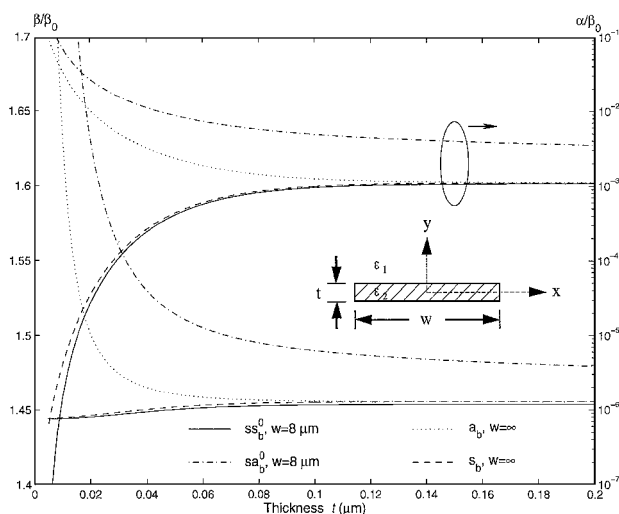


Fig. 1. Dispersion characteristics of the first two modes (ss_b^0 and sa_b^0) supported by a Au film of width $w = 8 \mu\text{m}$ embedded in an infinite background of SiO_2 . The dispersion curves of the s_b and a_b modes supported by the infinitely wide structure are shown for comparison. Values of $\epsilon_{r,1} = 2.085$ and $\epsilon_{r,2} = -131.9475 - j12.65$ were used for the SiO_2 and Au regions, respectively. The analysis was conducted at a free-space wavelength of 1.55 μm .

directions. The modes are not TM in nature; all six electromagnetic field components are present for all modes although the transverse electric field normal to the horizontal interfaces dominates for a structure with an aspect ratio $w/t > 1$. Four fundamental modes exist and are labeled following the nomenclature in Ref. 7: aa_b^0 , as_b^0 , sa_b^0 , and ss_b^0 . The letters a and s identify asymmetry or symmetry of the main electric field component along the y and x confinement directions, respectively, and b and l indicate a bound or a leaky mode, respectively.

The ss_b^0 mode is of most interest, since its attenuation diminishes and it evolves toward a bound quasi-TEM mode for decreasing metal thickness, in much the same way that the symmetric mode does in the infinitely wide metal film structure. Note, however, that the mode becomes less tightly bound as the guide becomes thinner. The dependence of the attenuation on metal thickness was illustrated by computation of the dispersion curves of a Au metal film embedded in an infinite SiO_2 background by use of the method of lines.^{7,8} The results are shown in Fig. 1 for infinitely wide and 8- μm -wide Au films. The curves are computed at a free-space wavelength of 1.55 μm and with values of $\epsilon_{r,2} = -131.9475 - j12.65$ for the Au film and $\epsilon_{r,1} = 2.085$ for the SiO_2 background.⁹ The attenuation curves of the ss_b^0 mode of the 8- μm structure can be seen to follow closely that of the film of infinite width.

The purpose of this Letter is to present, for what is believed to be the first time, experimental confirmation of the waveguiding abilities of thin metal films of finite width. To do so, qualitative and quantitative investigations of the dependence of these guides on the incident angle of polarization were undertaken. Details of the fabrication and experiments are presented, and an analysis of the results is given.

We designed and fabricated various laterally confined waveguide structures to confirm the theoretical analyses presented in Refs. 7 and 8. A 2- μm -layer of SiO_2 was deposited by a plasma-enhanced chemical-vapor-deposition system on a 7.62-cm-diameter 5- μm SiO_2 -on-Si wafer, for a total of 7 μm of SiO_2 . The 5 μm of native oxide was formed by standard thermal oxidation techniques. The 7- μm layer of SiO_2 is used as the bottom cladding for the waveguides being fabricated. The losses predicted by theory are related to the type of metal used, and noble metals such as Ag, Au, Al, and Cu were identified to be of significant interest. In previous work,^{1,5} Ag was used because its characteristics are best at the He-Ne wavelength. However, at optical communications wavelengths, the characteristics of Au are the most promising, in part because of the low waveguide losses predicted by theory. Consequently, the above fabrication process was followed by electron-beam deposition of 20 nm of Au. Waveguide structures of various widths were defined by use of standard lithographic techniques. Au deposition was followed by a second 2- μm layer of plasma-enhanced chemical-vapor-deposited SiO_2 . No annealing was performed at this time so that the absorption lines in the 1.5- μm band would be reduced. The sample was cleaved, and 20-nm-thick, 8- μm -

wide, 3.5-mm-long metal-film structures were obtained. These components were designed and fabricated to be consistent and to represent adequately the theoretical modal analysis and its boundary conditions are presented in Fig. 1.

To excite the ss_b^0 mode it is necessary to overlap the field distribution of this mode with the optical excitation signal as much as possible. The ss_b^0 mode, much like the s_b mode, is of interest because its field pattern lends itself to end-fire excitation.^{7,8} To match the dominating normal transverse electric field component, we used a highly polarized 1.55- μm semiconductor laser connected to a polarization-maintaining fiber to excite the input of an 8- μm -wide waveguide. The fiber was held in a rotation stage with a specially designed mount so that it could be physically rotated. By rotation of the fiber it was possible to study the polarization dependence of the waveguide. Figure 2 displays an infrared image of the optical mode at the output of the 8- μm waveguide. The effect of varying the polarization angle in 10° increments is clearly visible.

This experiment demonstrates the single-mode operation of the 8- μm -wide waveguide. As the position of the input fiber was shifted vertically or laterally, the intensity of the light that was recorded diminished, and no circularly symmetric spot could be observed. This

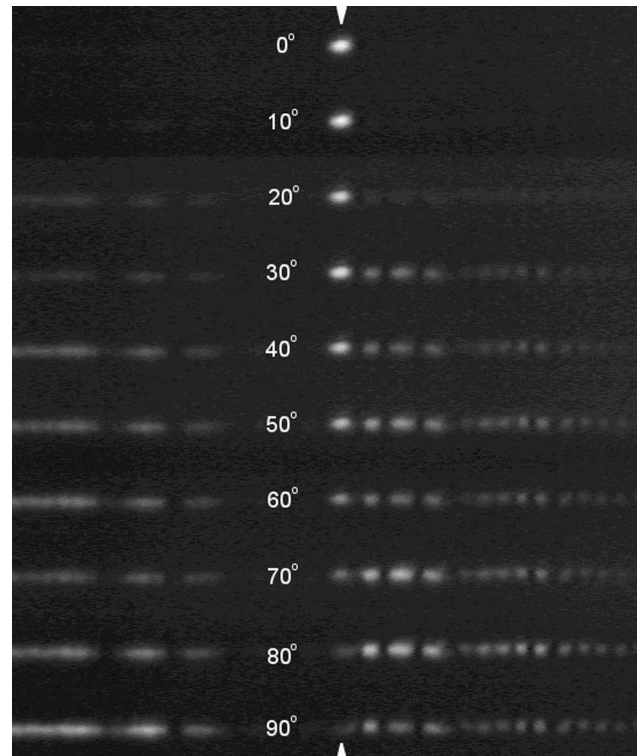


Fig. 2. Sequence of images of the output of an 8- μm -wide, 20-nm-thick, 3.5-mm-long Au waveguide for the indicated angles of incident polarization. A 0° angle corresponds to alignment of the electric field vector along the y axis of the waveguide structure—see Fig. 1. The aligned vertical arrows identify the location of the waveguide. The measurements were performed at a free-space wavelength of 1.55 μm .

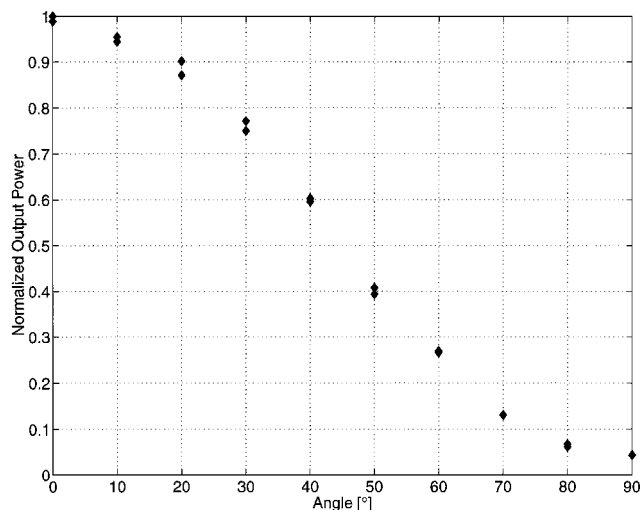


Fig. 3. Effect of polarization on the power transmitted through an 8- μm -wide, 20-nm-thick, 3.5-mm-long Au waveguide. A 0° angle corresponds to alignment of the electric field vector along the y axis of the waveguide structure—see Fig. 1. The measurements are performed at a free-space wavelength of 1.55 μm .

behavior is characteristic of laterally and vertically confined single-mode optical guides. A well-confined beam bound to the guide can readily be seen for 0° vertical polarization (electric field vector aligned along the y axis of the waveguide structure—see Fig. 1), as is expected from theory. Strong coupling for this structure was observed for polarization angles up to 30°. As the polarization angle is further increased, light is no longer efficiently coupled to the ss_b^0 mode. The optical excitation couples into slab modes along the structure and is seen as light propagating along each side of the metal waveguide. The asymmetry in the slab modes, as shown in Fig. 2, is attributed to slab reflections in the vicinity of other waveguides. As the polarization angle approaches 90° horizontal (electric field vector aligned along the x axis), the slab guiding dominates over any signal that is visible at the output of the 8- μm waveguide. It is apparent that very little coupling to the ss_b^0 mode exists in this condition. This qualitative investigation establishes the waveguiding abilities of thin finite-width metallic films and verifies the validity of some of the theoretical results presented in Refs. 7 and 8.

A series of polarization measurements were performed to strengthen the qualitative investigation of propagation presented above. Once again, the input polarization-maintaining fiber was physically rotated in 10° increments. A single-mode fiber was positioned at the output of the waveguide to collect the output radiated light. The polarization of the input light was initially set to be vertical for maximum coupling to the guide. With the use of an infrared camera and a microscope, we were able to ensure that the waveguide was properly excited by looking at the sample from

the top. When excited, the guide becomes illuminated and can easily be distinguished from adjacent features. This radiation is not the result of reflections off the metal and is associated with other waveguide losses and on the polarization of the input radiation.

With the guide properly excited, we aligned the input and output fibers to maximize the transmitted power. Once in position, the output fiber was not moved for the remainder of the experiment. The input fiber was rotated in increments of 10°, followed by alignment adjustments for a maximum measured transmitted power. We recorded measurements of the transmitted power for a polarization angle varying from 0° vertical to 90° horizontal in increments of 10° and back to 0° to verify repeatability. The resulting graph is shown in Fig. 3. The ordinate corresponds to the normalized power at the output of the 8- μm waveguide, and the abscissa corresponds to the polarization angle of the incident beam.

The results shown in Fig. 3 are consistent with the qualitative investigation shown in Fig. 2. The measured optical power throughput of the guide varies with the polarization angle in the same manner as was observed with the infrared camera. The transmitted power is highest for vertical polarization and decreases steadily until horizontal polarization is reached. Based on the theoretical results,⁸ it is expected that the polarization sensitivity of finite-width metal films would diminish for narrower waveguides.

What is believed to be the first experimental observation of the propagation characteristics of long-range plasmon-polariton waves guided by a thin metal film of finite width has been presented. The polarization dependence of the main mode of propagation has been investigated both qualitatively and quantitatively. The experimental results are in agreement with theoretical predictions, and the usefulness of the end-fire excitation technique for launching modes in this type of structure was demonstrated.

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